Total No. of Questions : 6]

SEAT No. :

P37

[Total No. of Pages : 2

Oct./TE/Insem.-151

T.E. (Civil)

HYDROLOGY AND WATER RESOURCE ENGINEERING

(2015 Pattern) (Semester - I)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates:

1) ***Answer Q1. or Q.2, Q.3 or Q.4, Q.5 or Q.6.***
2) ***Figures to the right indicate full marks.***

Q1) a) The mean annual rainfall in mm at '4' rain guage stations in catchment are 1050, 790, 700 and 660. If error in estimation of mean rainfall not exceeded 10%. Determine the additional number of rain guages needed. **[6]**

b) Explain with neat sketch A–V method of discharge measurement. **[4]**

OR

Q2) a) The total observed run off volume during a storm of 6 hr. duration with a uniform intensity of 15mm/hr is 21.6 Million m^3. If the catchment area is 300 km^2, find the average infiltration rate and the run off coefficient for the catchment. **[6]**

b) Which are the agencies that usually maintain rain guages and make observation in your state? At what time or times in day is rainfall recorded? **[4]**

Q3) a) The GCA of irrigation Canal is 1.20 lakh hectare. The CCA is 75% of GCA. Intensities of irrigation for Kharif and Rabbi crops are 40% and 55% respectively. If duties at canal are 800 and 1550 ha/cumecs for Kharif and Rabbi crops respectively. Determine the discharge at canal head. **[6]**

b) What are various methods of assessment of canal revenue. **[4]**

OR

P.T.O.

Q4) a) Find out capacity of reservoin from following data. The CCA = 80,000 ha. Take canal & reseroir losses 5% and 10% respectively. **[6]**

Crop	Base period (Days)	Duty (ha/cumecs)	Intensity of Irrigation (%)
Sugar Cane	320	2500	20
Rice	120	1800	25
Wheat	150	2000	30

b) Draw a labelled sketch of layout of drip irrigation system. Why is it essential in India? **[4]**

Q5) a) A well with diameter of 1.2m penetrates an unconfined aquifer of thickness 70 m & coefficient of permeability (K) = 50 m/day. The well is pumped so that water level in the well remains at 60 m above the datum. Water table is at 500 m.

Determine the discharge in litres ? **[5]**

b) Define Aquifen and describe various types of aquifer. **[5]**

OR

Q6) a) In confined aquifer of 10 m thick, 12 cm diameter well is pumped at rate of 250 litres/min. Drawdown observed in two wells located at 11m and 48m distance from centre of well are 4m and 0.09m. Determine the coefficient of permeability and transmissibility? **[5]**

b) What is Darcy's law. State its assumptions and limitations in ground water. **[5]**

⌘⌘⌘